



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	B.TECH	IIT Kharagpur	8.24 / 10
2023	Class XII, GSEB	Ashadeep Ucchatar Madhyamic Shala	80.90%
2021	Class X, GSEB	Ashadeep International School	90.00%

INTERNSHIPS

Work2Hire | Founder's Office Intern

May'26 - Aug'26

- Lifted parse success on multi-column resumes from **71% to 92%**, rebuilding the ingestion pipeline around word-box geometry and x-coordinate clustering
- Halved LLM cost per resume at unchanged **0.89** extraction accuracy, by resolving common fields from **gazetteer** rules and cache before the constrained call
- Shipped the ATS scorer as a **versioned REST API** returning an explainable per-section breakdown, ranking **BM25** term coverage against each job description

Digikestra | Software Development Intern

Dec'25 - Mar'26

- Ended duplicate orders and cut failed payment transactions by **35%**, keying API writes on **idempotency tokens** so retries deduplicate and never double-post
- Tripled page speed across the six busiest pages, cutting p95 from **2.1s to 0.7s** by collapsing **N+1 queries** and adding the composite indexes **EXPLAIN** called for
- Lowered bugs reaching production from **4 to 1 per release**, adding **48 boundary-value regression tests** over 7 modules, each pinned to the root cause of a bug

AnTim Technologies | Technology Associate

Aug'25 - Nov'25

- Generated **7 in 10** modules ready to merge unedited, using a **schema-constrained LLM** to turn a client spec into typed JSON that code templates build from
- Consolidated the **30%** of code duplicated across 12 past client projects into 5 reusable **Angular** and **.NET** libraries, removing repeat work from new builds
- Shortened first delivery on new projects from 9 days to 4, gating CI on the **contract tests** every generated module had to compile and pass before review

PROJECTS

Pathfinder Agent | can an LLM replace a real graph solver? | RAG, pgvector, NetworkX, Python

Apr'26 - May'26

- Surfaced a circular dependency in Apache Airflow using **Kahn, iterative Tarjan, Dijkstra and A*** over a live **411-package** PyPI graph, cross-checked with **NetworkX**
- Benchmarked the **RAG agent** against those exact solvers and ruled out the **context truncation** theory: all 19 failing cases fit inside **512 tokens**, explaining none
- Correlated plan validity with prompt length at $r = -0.79$, and then reshaped the **tool output format** to lift validity from **58% to 80%** against the solver as truth

Forge Pipeline | can a model ship like code, and where does it break? | PyTorch, TorchScript, GitHub Actions

Mar'26

- Automated retraining and deployment of a **468K-parameter CNN** to **93.4%** test accuracy inside **GitHub Actions**, stamping every build with its **dataset fingerprint**
- Gated every release on the **TorchScript** export matching **PyTorch** to **5 decimal places** across 512 samples, with **SHA-256** artefact checks and a one-call rollback
- Broke the endpoint deliberately to find its ceiling: **p99** crossed the 100ms mark at **1,100+ requests/s**, where **Little's Law** held to **0.04%**, confirming saturation

Sentinel Gateway | how much speed does dropping to C++ actually buy? | C++17, pybind11, FastAPI, Java

Jan'26 - Feb'26

- Accelerated the rate limiter **2.45x** over Python in **C++17/pybind11**, then found per-call **GIL** release ran 12x slower than holding it and batched to recover that
- Fronted **Python** and **Java services** behind **per-tier quotas** and **circuit breakers**, correctly shedding the 45 over-quota requests inside a single 60-request burst
- Cut false positives to zero with a **MAD-based robust z-score** detector, scoring over a sustained window rather than per request, widening that separation **6x**

SKILLS AND EXPERTISE

Languages: Python, C++, Java, JavaScript, TypeScript, SQL | **CS:** data structures and algorithms, system design, OOP, concurrency, testing, debugging, Git
Backend, Cloud and DevOps: FastAPI, REST APIs, microservices, PostgreSQL, pgvector, Docker, Kubernetes, AWS, Azure, GCP, Linux, CI/CD (GitHub Actions)
Machine Learning and Generative AI: PyTorch, TensorFlow, scikit-learn, pandas, deep learning, NLP, computer vision, LLMs, RAG, embeddings, AI agents
Coursework: Artificial Intelligence: Foundations and Applications, AI/ML for Robot Autonomy, Programming and Data Structures (Theory and Laboratory)

ENTREPRENEURIAL EXPERIENCES

Co-Founder | TEVIX (SensoPack) | 3rd in Agriculture track, Platinum Jubilee Innovation Challenge, INR 5L funding commitment

Apr'26 - Aug'26

- Built **SensoPack**, an AI-IoT cold-chain tag combining biodegradable biofilms, biomarker sensing and on-device **TinyML** on an **INR 30-120** NFC bill of materials
- Achieved **92%** predictive shelf-life accuracy on the tag, replacing static printed expiry dates with live freshness estimates and projecting a **30%** cut in waste
- Addressed an **INR 61,000 crore** fisheries market and **INR 30,000 crore** of annual livestock losses, making freshness uncertainty the wedge into fresh retailers

COMPETITION/CONFERENCE

Bronze Medal, OpenSoft | Inter-Hall General Championship | React, Zustand, WebSockets

Mar'26

- Sustained **80+ trades per second** in a **real-time trading UI**, cutting **React** re-renders with narrow **Zustand** selectors under a continuous live market update load
- Merged **REST** history with buffered **WebSocket** streams into one continuous series, so that live candles extended the chart seamlessly on **Lightweight Charts**
- Eliminated race conditions across multi-store synchronisation under sustained load using the **Zustand subscribeWithSelector** middleware and scoped reads

Inter IIT Tech Meet 14.0, Adobe | Reimagining Photoshop as the AI editor of 2030

Oct'25 - Dec'25

- Enabled earlier edits to be re-tuned without regenerating the image, by caching per-instruction **velocity vectors** and linearly interpolating them at **inference**
- Chained **SAM 3 segmentation**, 2% mask dilation and edge-shadow adjustment, **LaMa inpainting** and **FLUX.2** recomposition into one-gesture drag repositioning
- Selected **FLUX.2** over **FLUX.1** and **SDXL** on prompt adherence, and **Qwen3-VL-8B** over **CLIP** and a 337-class classifier on hand-stroke and attribute tracking too

POSITIONS OF RESPONSIBILITY

Public Relations Chair | Technology Students' Gymkhana, IIT Kharagpur

Jun'25 - Jun'26

- Orchestrated a **5-day** round-the-clock SIH Hardware Edition on an **INR 1.2 crore** budget, coordinating **450+** participants, mentors, volunteers and vendors
- Launched the **SoAC Portal** for 70+ student societies, centralising discovery; drove recruitment and event updates for 16,000+ students and 30,000+ alumni
- Secured **40+** institutional collaborations and sessions, sharing **1,000+** internship and career openings; grew TSG social reach **30% YoY** to 40,000+ followers